

Boxplots with ggplot2

[Boxplots](#) are a convenient way of graphically depicting groups of numerical data through their quartiles. Box plots may also have lines extending vertically from the boxes (whiskers) indicating variability outside the upper and lower quartiles, hence the terms box-and-whisker plot and box-and-whisker diagram. Outliers may be plotted as individual points.

Let's see how we can create them with `qplot` and `ggplot2`!

In [3]:

```
library(ggplot2)
df <- mtcars
```

In [4]:

```
head(df)
```

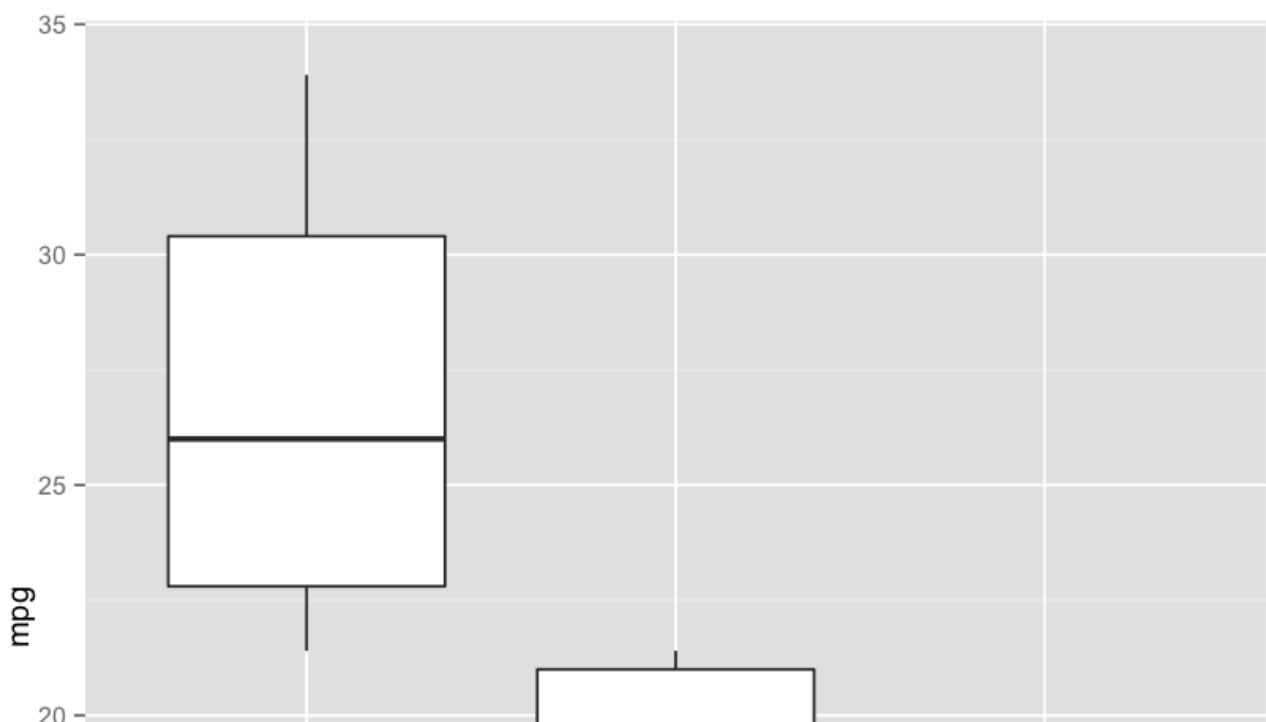
Out[4]:

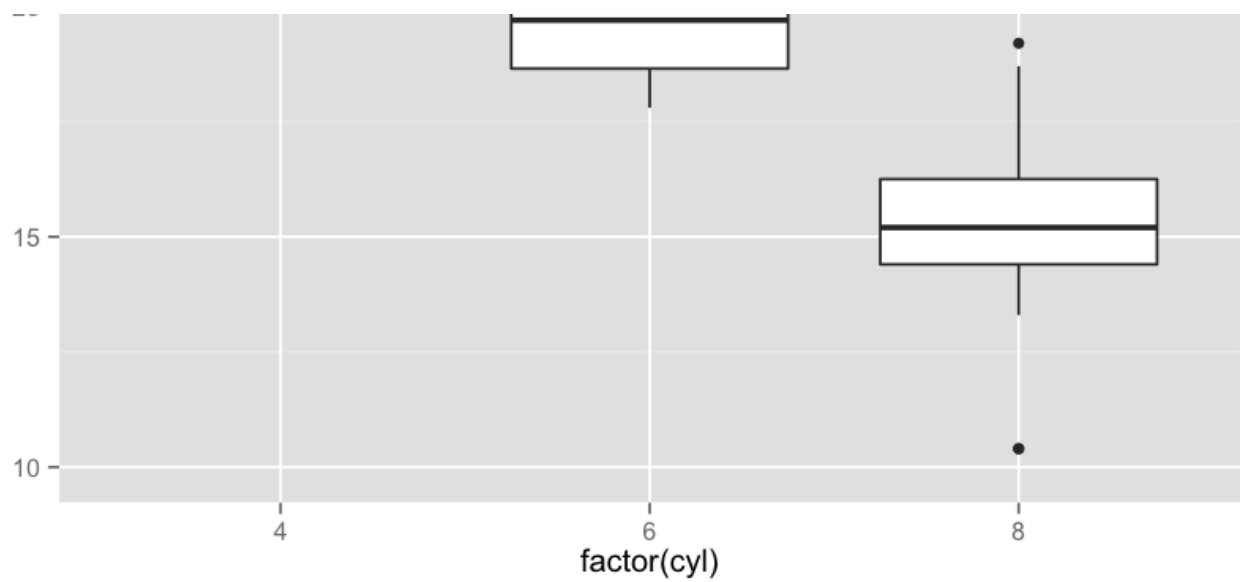
	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21	6	160	110	3.9	2.62	16.46	0	1	4	4
Mazda RX4 Wag	21	6	160	110	3.9	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.32	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.44	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.46	20.22	1	0	3	1

qplot

In [10]:

```
qplot(factor(cyl), mpg, data = mtcars, geom = "boxplot")
```

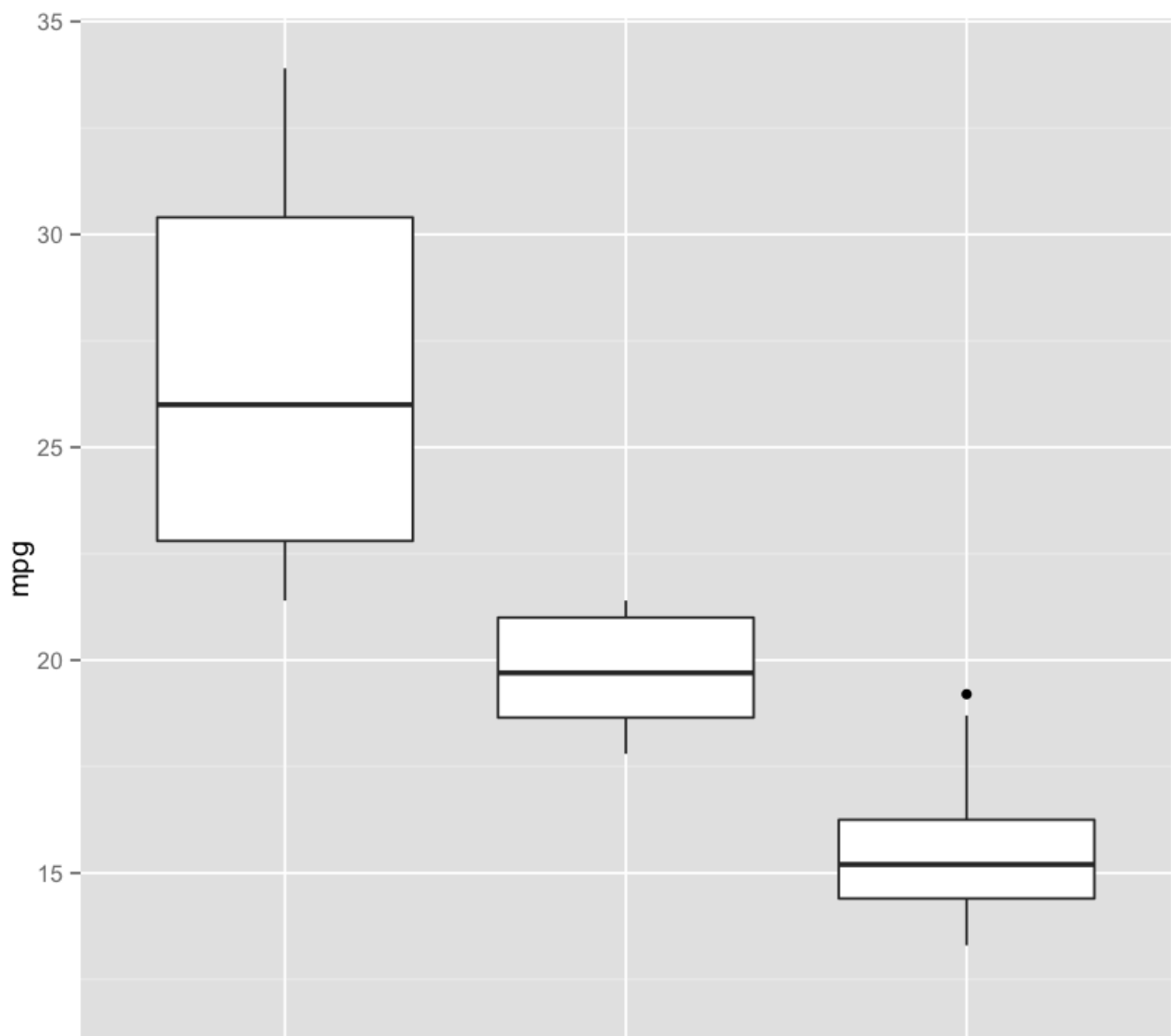


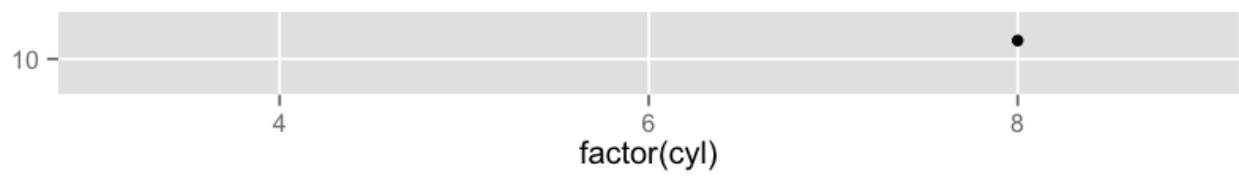


ggplot

In [11]:

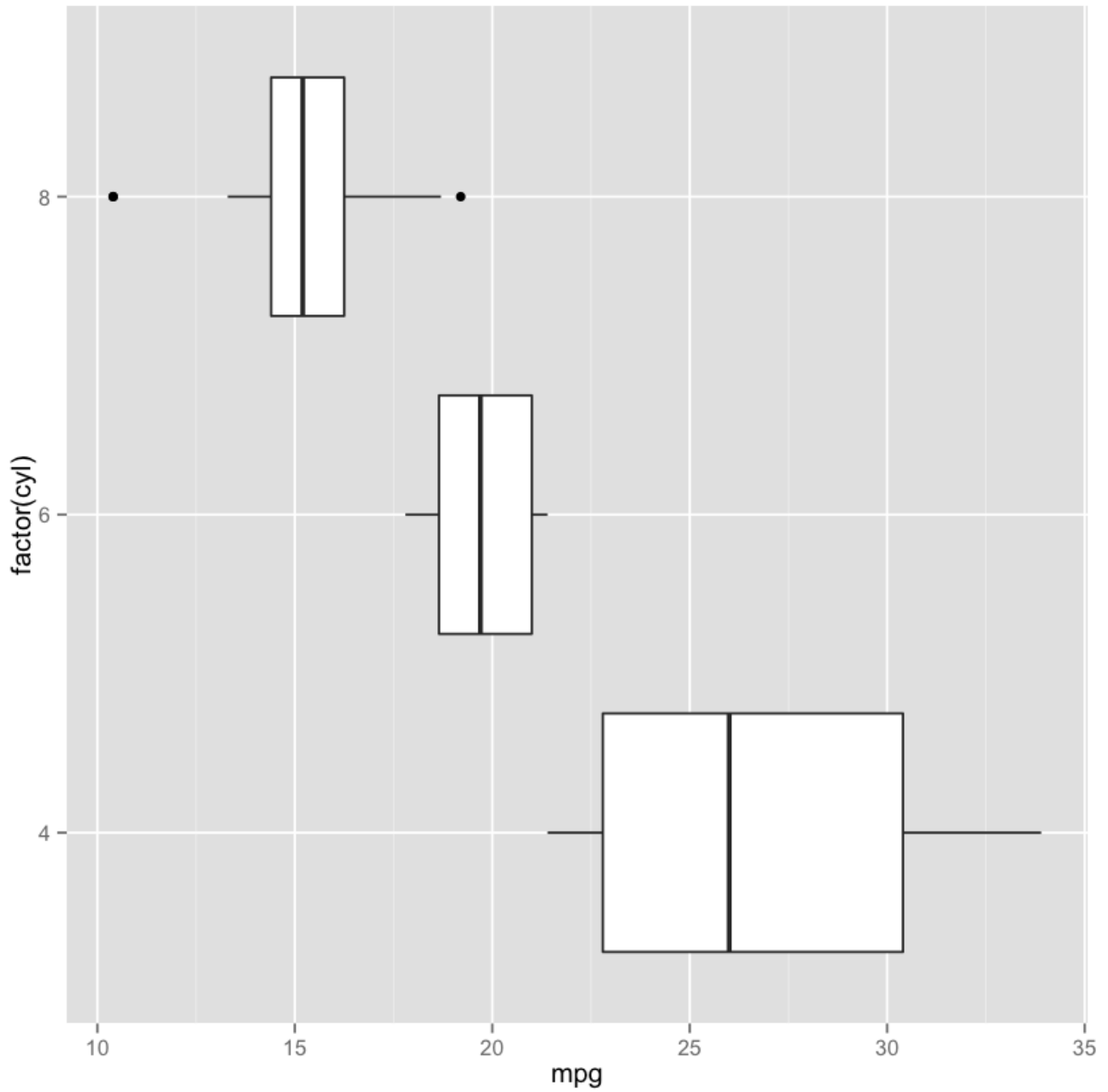
```
pl <- ggplot(mtcars, aes(factor(cyl), mpg))  
pl + geom_boxplot()
```





In [13]:

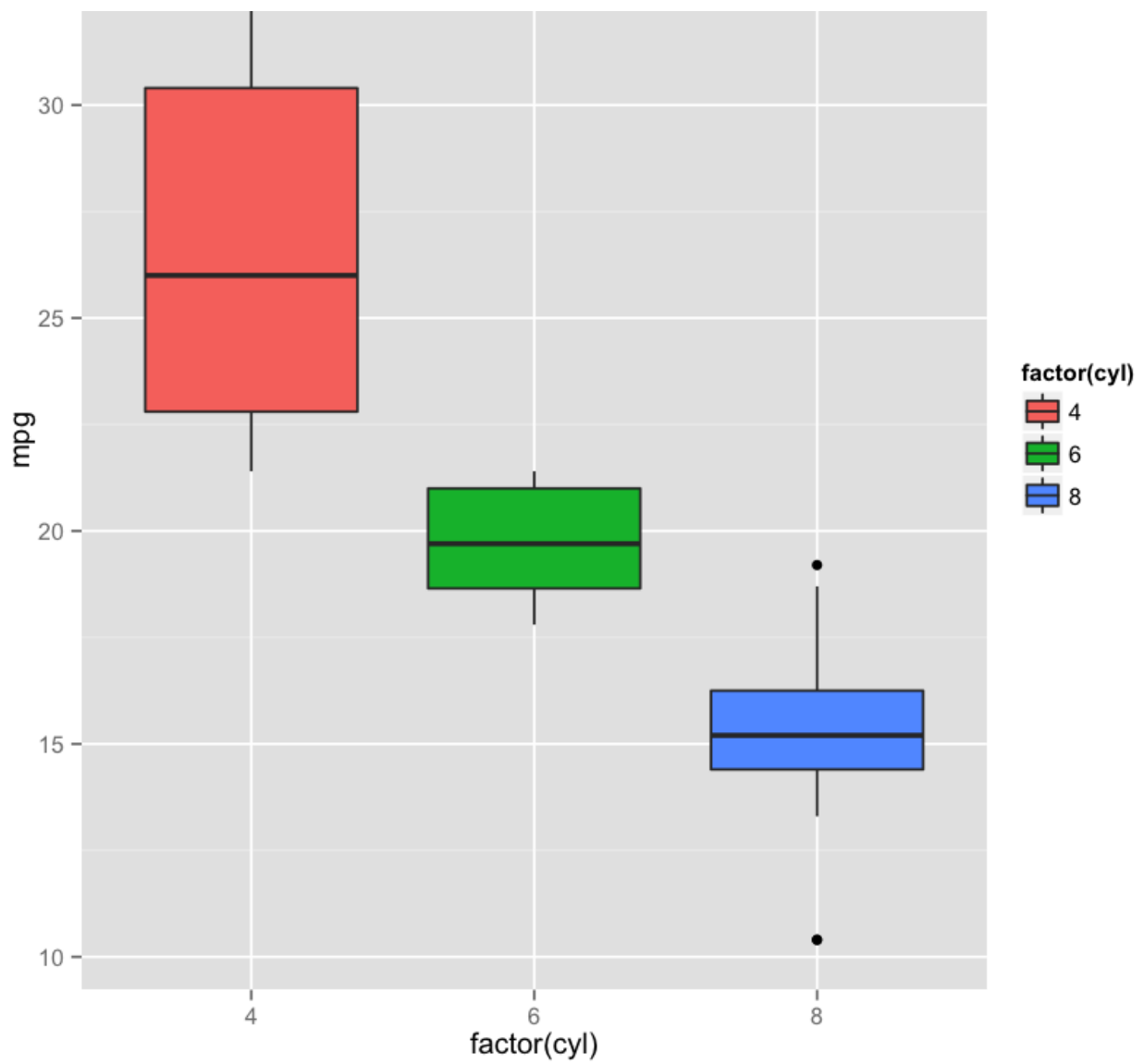
```
pl + geom_boxplot() + coord_flip()
```



In [14]:

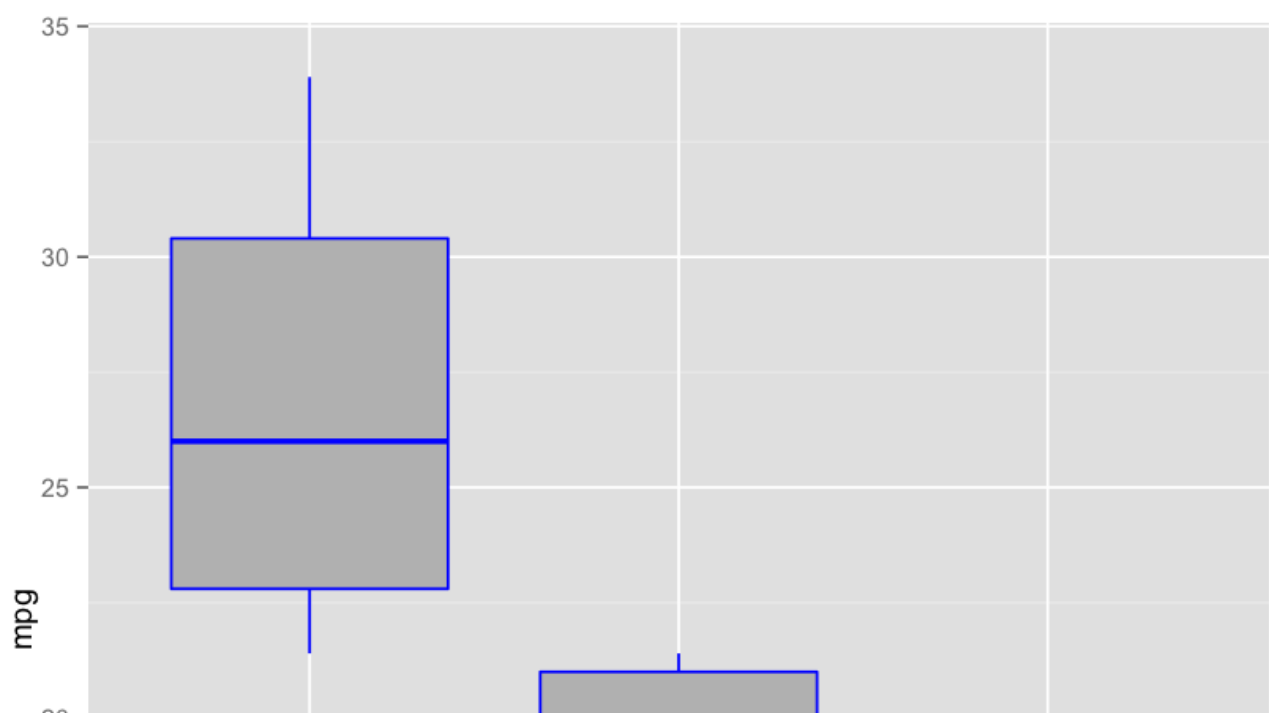
```
pl + geom_boxplot(aes(fill = factor(cyl)))
```

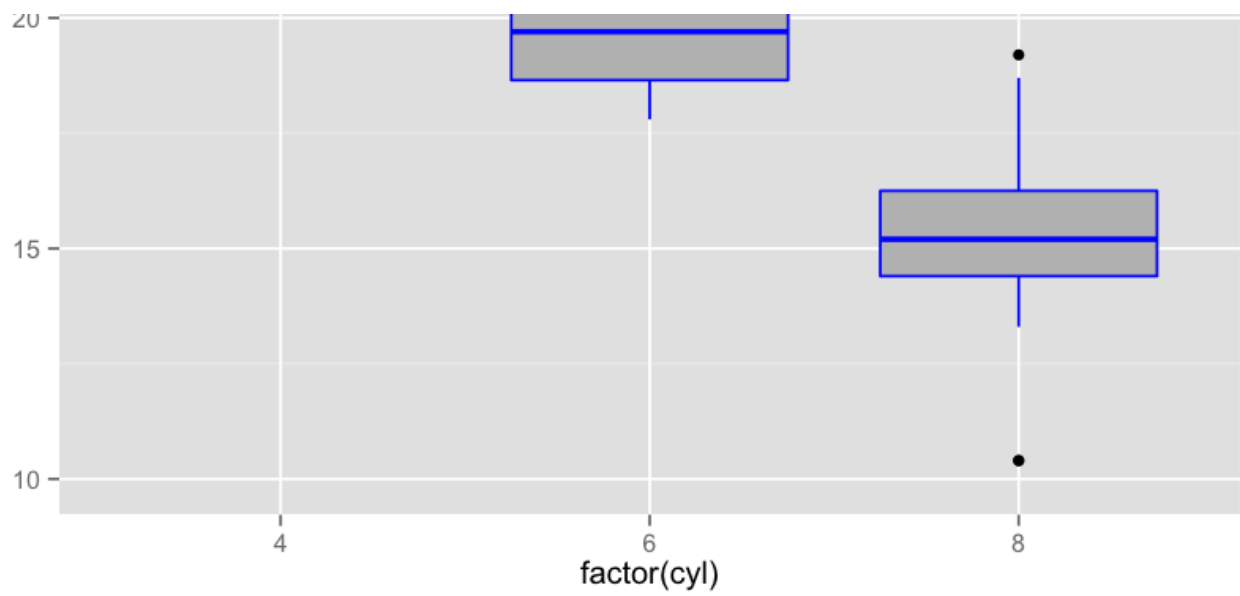




In [16]:

```
pl + geom_boxplot(fill = "grey", color = "blue")
```





That's it for the basics of creating boxplots in ggplot2!